As for the normal phase, the choice will, to start with, be the simplest possible one: each path is either possible or not, corresponding to the probability weights 1 and 0. During the extreme phases, this assumption is no longer reasonable. Again the model will be extremely simplified, but still it is based on physical intuition and, most importantly, completely time symmetric. Assume that the only types of edges having a non-neglectable chance of occurring during the extreme phase $[-m-1, -m]$ are of the following two kinds: The first scenario is that the universe passes through the extreme phase into a state of zero entropy. The other scenario is that it passes into a state with high entropy (equal to $2m$). Universes of one of these two types will be given the (un-normalized) probability 1 or p, respectively. Here $p > 0$ should be thought of as a very small number, at least when the size of the model becomes large. During the other extreme phase $[m, m+1]$, near the Big Crunch, we make the completely symmetric assumption.

Remark 3. These assumptions may perhaps seem somewhat arbitrary. And to a certain extent, this may be so. However, they do represent the following viewpoint of what may happen at the full cosmological scale: we may think of the Big Bang and the Big Crunch as states of complete order with zero volume and entropy. Such states can very well be metastable, very much like an oversaturated gas at a temperature below the point of condensation. If no disturbance takes place, such metastable states can very well continue to exist for a substantial period of time. In particular, a low-entropy state can have a very good chance of surviving the intense but extremely short extreme phase. On the other hand, if a sufficiently large disturbance occurs, then the metastable state may almost immediately decay into a very disordered state of high entropy.

It is not my intension to further argue in favor of this viewpoint here. The main thing in this chapter is to show that completely symmetric boundary conditions at the endpoints may give rise to a broken time symmetry.

The multiverse now splits up into four different kinds of paths:

- LL: The entropy is low (=0) at both ends ($-m$ and m).
- LH: The entropy is 0 at $-m$ and $2m$ at m.
- HL: The entropy is $2m$ at $-m$ and 0 at m.
- HH: The entropy is high ($= 2m$) at both ends ($-m$ and m).

If we now denote by N_{LL}, N_{LH}, N_{HL} and N_{HH} the number of paths of the indicated kinds, then with the above assumptions we also get the corresponding probability weights for the corresponding types as

$$P_{LL} = N_{LL}, \quad P_{LH} = pN_{LH}, \quad P_{HL} = pN_{HL}, \quad P_{HH} = p^2 N_{HH}. \tag{10}$$

We can now consider the following two types of broken time symmetry:

Definition 4. A multiverse is said to exhibit a *weak* broken time symmetry if

$$P_{LL} \ll P_{LH} + P_{HL}. \tag{11}$$

Definition 5. A multiverse is said to exhibit a *strong* broken time symmetry if

$$P_{LL} + P_{HH} \ll P_{LH} + P_{HL}. \tag{12}$$

Both these definitions should of course be made more precise when applied to specific models for the multiverse, e.g., by showing that the corresponding limits